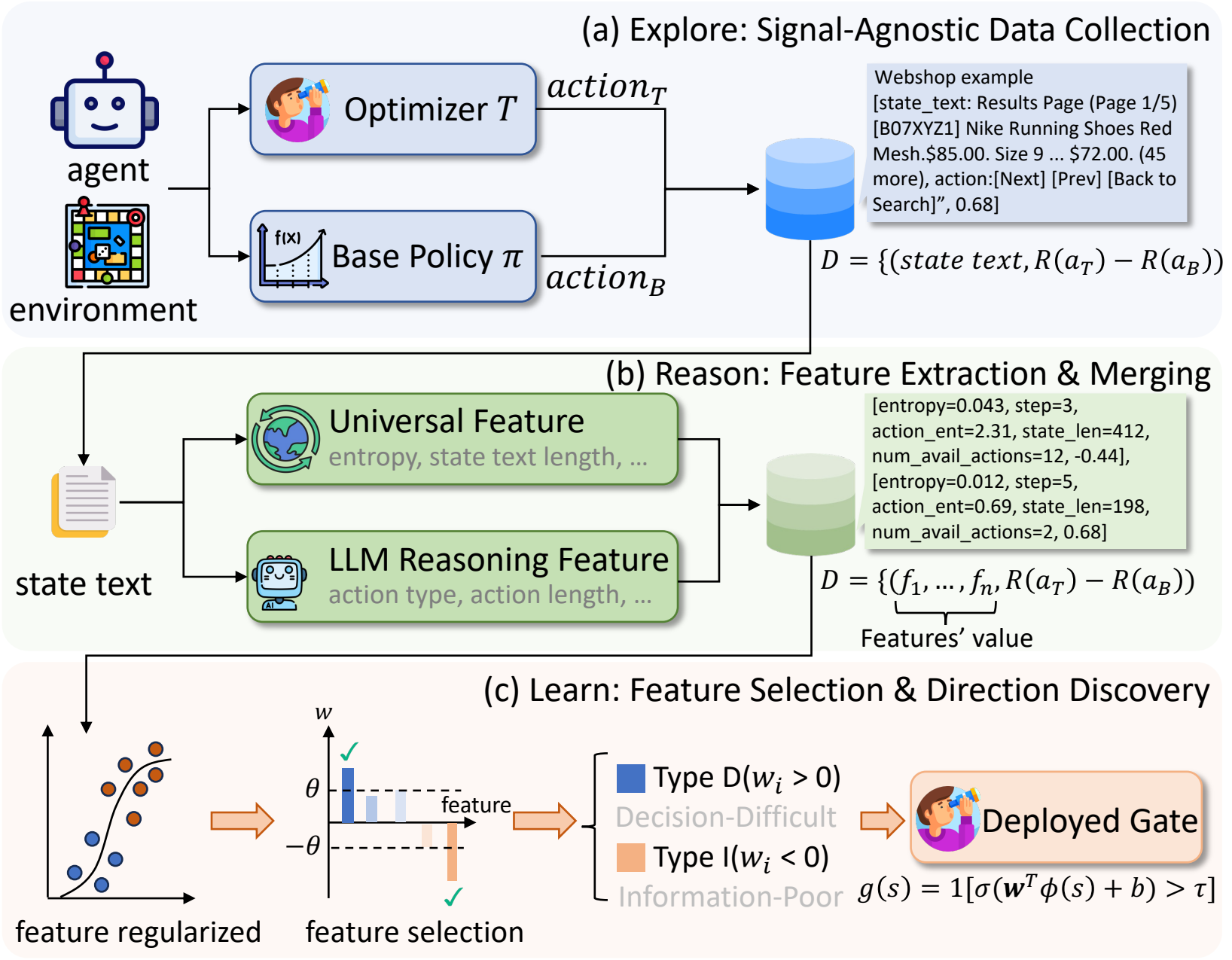
(a) Explore: Signal-Agnostic Data Collection
agent
environment
Optimizer $T$
Base Policy $\pi$
$action_T$
$action_B$
Webshop example
[state_text: Results Page (Page 1/5) [B07XYZ1] Nike Running Shoes Red Mesh.$85.00. Size 9 ... $72.00. (45 more), action:[Next] [Prev] [Back to Search]", 0.68]
$D = \{(state\ text, R(a_T) - R(a_B))$
(b) Reason: Feature Extraction & Merging
state text
Universal Feature
entropy, state text length, ...
LLM Reasoning Feature
action type, action length, ...
[entropy=0.043, step=3, action_ent=2.31, state_len=412, num_avail_actions=12, -0.44], [entropy=0.012, step=5, action_ent=0.69, state_len=198, num_avail_actions=2, 0.68]
$D = \{(f_1, \ldots, f_n, R(a_T) - R(a_B))$
Features' value
(c) Learn: Feature Selection & Direction Discovery
feature regularized
$w$
$\theta$
$-\theta$
feature
feature selection
Type D($w_i > 0$)
Decision-Difficult
Type I($w_i < 0$)
Information-Poor
Deployed Gate
$g(s) = 1[\sigma(\boldsymbol{w}^T \phi(s) + b) > \tau]$